ColonAI

A second pair of eyes for colon-cancer screening.

ColonAI is a deployment-ready, patient-friendly multimodal AI for colon-cancer screening. It looks at a colonoscopy image, the patient's symptoms, and their medical history together — and refuses to give a confident reading when its internal safety checks say it shouldn't.

+136%	0.61	21 FPS	3
improvement on the scope brand the model	cross-vendor segmentation IoU	live-video throughput on a basic MacBook	independent safety layers — abstains

Why this exists

About 1 in 4 polyps is missed during routine colonoscopy. Existing AI helpers have a dangerous failure mode: when a hospital uses a different scope brand than the model was trained on, the AI still says “polyp” with high confidence — but the highlight on the screen lands on the scope's display overlay, not on the actual lesion. The doctor sees the green light and trusts it. We measured this directly with the original model on Pentax scopes:

■ Classified **95%** of Pentax polyps as polyps. ■ Heat-map landed in the wrong place **93%** of the time.

A model that's right for the wrong reason is one of the most dangerous failure modes in medical AI. ColonAI fixes it.

What ColonAI does in one sentence

It gives the doctor a confident answer when it can, a plain-English *“please ask a human”* when it can't, and refuses to look at uploads that aren't actually colonoscopy images — backed by three independent safety layers and validated on five external scope datasets.

How a clinician (or patient) uses it

1. Open the URL

No install, no signup, no payment. huggingface.co/spaces/Yuvraj2319/colonai — works on any browser, any device.

2. Patient info

Plain form: name, age, BMI, smoking, alcohol, family history. A one-click **accessibility toggle** in the sidebar enables larger fonts, dyslexia-friendly typography, and big tap targets for older users.

3. Symptoms

Free-text symptoms ("rectal bleeding for two weeks") plus a few fast-track red-flag checkboxes.

4. Upload the image

Drag-and-drop a colonoscopy frame. **If it's not really an endoscopy frame, the system refuses on the spot** — no fake result.

5. Analyse

In 1–2 seconds, six specialised agents run in sequence: image, symptom-text, patient-history, fusion, explainability, clinical recommendation. A cross-check confirms they all agree before showing the answer.

6. Results page

Plain-English narrative + heat-map overlay + "why this answer" rationale + per-modality contribution chart + clinical recommendation aligned with NICE NG12 / BSG guidelines + downloadable PDF report.

7. Live Video Mode (optional)

For real-time colonoscopy. Runs at **21 FPS on a basic MacBook**. A single-frame false positive never reaches the screen — a polyp must persist across 3 consecutive frames before being flagged.

The six AI agents working in concert

#	Agent	What it produces
1	Image agent	Class probability + GradCAM++ heat-map showing what the AI was looking at
2	Symptom-text agent	Reads symptom text with a clinical NLP model; extracts medical concepts
3	Patient-history agent	Risk score from age / BMI / family history / smoking / alcohol
4	Fusion agent	Combines all three modalities with learned weights showing what dominated
5	Explainability agent	MC-Dropout uncertainty + Integrated Gradients (a second independent attribution)
6	Recommendation agent	Urgency + surveillance interval, mapped to NICE NG12 / BSG guidelines

Three independent safety layers

ColonAI doesn't trust any single signal. A finding only reaches the user if it passes through three completely separate checks. **In screening, silence is safer than overconfidence.**

Layer 1 — Symptom-driven safety net (pure clinical rules)

A rule engine reads the symptoms the patient typed. If they match the NICE NG12 fast-track criteria (rectal bleeding ≥ 50 yr, iron-deficiency anaemia ≥ 60 yr, weight loss + abdominal pain, severe-pain combinations), urgency is **escalated regardless** of what the AI says. Rules can only escalate, never down-grade.

Layer 2 — Image-statistics safety net (independent of the model)

A separate pixel-statistics engine looks for visible signs the trained model wasn't built to recognise: heavy bleeding, deep cavitation, very disordered edges, unusual colour distributions. If the picture is abnormal in a way the model can't explain, the system flags it for review even if the model is confident.

Layer 3 — Cross-agent consistency net

The pathology classifier, the GradCAM heat-map, a dedicated **segmentation decoder** (a separate neural network that draws a polyp mask), and **Integrated Gradients** (a second attribution method) all have to **agree** on what they're seeing. The system computes a coherence score as the geometric mean of three pairwise checks; if any single signal is low, coherence collapses and the system abstains.

Action	When it fires	What the user sees
■ SHOW	All checks passed	Full result: heat-map, plain-English explanation, recommendation
■ ABSTAIN	Confidence < 0.75 / uncertainty > 0.30 / diffuse heat-map / agents disagree	“Please ask a doctor to review this. The AI isn't confident enough.”
■ REJECT	Not a colonoscopy image / upload too large / pipeline error	“We can't analyse this image. Upload a real endoscopy frame.”

Every prediction is audited. The image's SHA-256 hash, the verdict, the confidence, and the flags are written to a permission-protected log file for post-hoc review. Hospitals can comply with their internal audit requirements out of the box.

Honest numbers — cross-vendor held-out tests

We do **not** report a single accuracy number — that's the metric most easily gamed. Here are the real per-vendor numbers, measured on five external datasets the model never saw during training.

Heat-map quality (Intersection-over-Union vs ground-truth polyp mask)

Dataset	Vendor	Original model	ColonAI v2	Lift
Kvasir-SEG	Olympus	0.24	**0.42**	**+79%**
ETIS-LaribPolypDB	Pentax (unseen)	0.07	**0.16**	**+136%**
CVC-ColonDB	Olympus	0.11	**0.21**	**+88%**
CVC-300	Olympus	0.12	**0.13**	**+14%**
Kvasir-test	Olympus	0.21	**0.42**	**+95%**

The Pentax row (red) is the key result. The original model classified Pentax polyps correctly but the heat-map landed in the wrong place. ColonAI v2 more than doubled the heat-map accuracy on Pentax.

Segmentation — a dedicated polyp-mask decoder (vendor-bias resolved)

Dataset	Vendor	Segmentation IoU	Detection sensitivity @ IoU≥0.5	mAP
Kvasir-SEG	Olympus	0.62	0.75	0.64
ETIS-Larib	Pentax	0.61	0.38	0.36
CVC-ColonDB	Olympus	0.62	0.64	0.54
CVC-300	Olympus	0.62	0.92	0.87
Kvasir-test	Olympus	0.62	0.73	0.64

The vendor gap is gone for segmentation. Pentax and Olympus get essentially identical IoU (0.61–0.62). This is what makes ColonAI deployable in a hospital that uses any scope brand, not just the one it was trained on.

Calibration — the 76% confidence really means 76% accurate

Expected Calibration Error (ECE) = **0.062** on a held-out validation set, after temperature scaling. Translation: when the system says it is 76% confident, it is correct ~76% of the time on similar cases. Most published medical AI reports ECE between 0.05 and 0.15.

Why ColonAI beats off-the-shelf colonoscopy AIs

Most colonoscopy AIs in the literature report a single accuracy number on a single dataset. ColonAI is different in five specific, measurable ways.

What	Most published AIs	ColonAI
Metrics reported	1 (accuracy)	5 per-vendor: acc, F1, IoU, ECE, FPPI
Cross-vendor validation	Not reported	5 datasets incl. Pentax
Calibration	Not reported	ECE = 0.062, temperature-scaled
Abstains when uncertain	Always gives an answer	3-mode policy: show / abstain / reject
Explainability methods	GradCAM only	GradCAM + Integrated Gradients + segmentation
Out-of-domain rejection	None	Pixel-stats endoscopy gate
Live-video debouncing	Per-frame flicker	3-frame persistence required
Audit trail	None	SHA-256-keyed log of every prediction
Patient-friendly UI	Clinical-jargon-only	Plain-English + accessibility mode
Security hardening	None	Upload validator, XSS guards, API auth, CORS
Deployment	Research code only	Live Space + Docker + REST API

The single line that sums it up

> *“ColonAI is a colonoscopy AI that refuses to be wrong-but-confident. It tells you what it sees in everyday language, shows you exactly where it's looking, says when it's not sure, and rejects what it shouldn't be looking at — backed by three independent safety layers and measured on five different scope brands.”*

How to evaluate it and deploy it

Try it right now (30 seconds, no setup)

Open **huggingface.co/spaces/Yuvraj2319/colonai** in any browser. Click *"Load Case A · Sigmoid Polyp"* on the Patient Info page, then *"Analyse"*. You'll see the full pipeline run in under 30 seconds.

Three ways to deploy in your environment

Option	Where	Effort	Best for
Hugging Face Spaces	Free public hosting	5 min, 4 clicks	Demos, pilots, academic review
Streamlit Cloud	Free public hosting	5 min, GitHub link	Demos, pilots
Self-host (Docker)	Your own server	Dockerfile included	Production / hospital deployment behind a firewall
REST API	Your hospital systems	FastAPI service	Integration with PACS, EHR, scope software

Security & compliance — out of the box

- Upload-size cap (10 MB), MIME/extension allow-list, magic-byte check, decompression-bomb guard.
- Defaults to localhost-only; refuses to bind to public interfaces without an API key.
- X-API-Key header authentication on every endpoint when configured.
- CORS allow-list configurable, defaults to safe values.
- Streamlit CSRF protection enabled; HTML escape on every user-typed string.
- Audit log written to a permission-restricted file (owner-only).
- Sanitised error responses — clients see a request-ID, the stack trace stays in the server log.
- Tracked, well-documented threat model in **SECURITY.md**.

Datasets used (all publicly available, research-licensed)

HyperKvasir (Norway) · CVC-ClinicDB / CVC-ColonDB / CVC-300 (Barcelona) · Kvasir-SEG · ETIS-LaribPolypDB (Pentax) · TCGA clinical metadata. **No patient-identifiable data is used. No private hospital data is used.**

Honest limitations (read this carefully)

Pretending a medical-AI system is perfect is dangerous. Here is the full list of things ColonAI does NOT do well, and the mitigations.

- **UC severity grading is the weakest sub-task.**

  Moderate-severe ulcerative colitis recall is only 0.15 — the model tends to over-call "mild." The cross-check safety net flags low-coherence cases for human review.

- **Cross-vendor per-polyp detection (Pentax) is still weaker than within-vendor.**

  Per-polyp F1 @ IoU = 0.5 is 0.38 on Pentax vs 0.75 on Olympus. Segmentation is at parity (0.61); per-polyp detection is not yet. Safety net catches uncertain cases.

- **Highly atypical lesions and invasive cancers are out-of-distribution.**

  The training corpus is screening-stage only (no Stage III/IV tumours, no fungating masses). These are flagged by the image-statistics safety net and the system asks for human review — ColonAI cannot classify them.

- **Confidence calibration is on the public-data validation distribution.**

  Per-site re-calibration is needed before clinical use in a specific hospital. We provide the calibration script.

- **Live-video FPS depends on hardware.**

  21 FPS on Apple Silicon (basic MacBook); a GPU is needed for 4K-resolution feeds at 30+ FPS. CPU-only deployment is fine for HD.

These limitations are exactly why the three-layer safety net exists. The combination of "best-effort prediction" + "abstain on uncertainty" + "reject on out-of-domain" + "audit every prediction" is what makes ColonAI safe to deploy in a screening setting.

Next steps

1. **Try the live demo** — huggingface.co/spaces/Yuvraj2319/colonai. Upload your own colonoscopy frames or use the built-in demo cases.
2. **Read the source** — github.com/Yuvraj235/Agentic_Multimodal_Colon_Cancer_AI. Everything is open, including the safety policy and threat model.
3. **Talk to the author** about pilot integration, per-site re-calibration, or a hospital-grade Docker deployment with hardware-backed TLS.

Yuvraj Pratap Singh
M.Sc. dissertation project — Amity University
GitHub: **github.com/Yuvraj235** · Hugging Face: **huggingface.co/Yuvraj2319**
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